k-Combinations

- **Example**
  In how many ways can we select three students to form a study-team, from a group of four students?
- Note: *the order in which the students are selected does not matter*!
- **Solution:**
  We need only to find the number of subsets containing 3 elements from the set of 4 students. There are 4 such subsets (each subset excludes one student). Therefore, there are 4 ways to choose the three students for the team.
- Equivalent Question: How many bit strings of length four have exactly three 1?